

Stat 122 (Mathematical Statistics II)

Lesson 3.5. Small Sample Confidence Intervals

Learning Objectives

At the end of this lesson, students are expected to construct and interpret confidence intervals for parameters of common distributions based on small samples.

Introduction

We have already discussed how one can use large-sample arguments to justify the use of large-sample confidence intervals like

$$\bar{Y} \pm z_{\alpha/2} \left(\frac{S}{\sqrt{n}} \right)$$

for estimating a single population mean, μ , and

$$(\bar{Y}_1 - \bar{Y}_2) \pm z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

for estimating the difference of two population means, $\mu_1 - \mu_2$.

The immediate question now is: What happens if the sample size n (or the sample sizes n_1 and n_2 in the two-sample case) is/are not large? How appropriate are these intervals?

Unfortunately, neither of these confidence intervals is preferred when dealing with small sample sizes. Thus, we need to treat small-sample problems differently. In doing so, we will assume (at least initially) that we are dealing with normally distributed data.

One population mean

Suppose that $Y_1, Y_2, \dots, Y_n$ is an *iid* sample from a $N(\mu, \sigma_0^2)$ population. If σ_0^2 is known, we have already seen that

$$\bar{Y} \pm z_{\alpha/2} \left(\frac{\sigma_0}{\sqrt{n}} \right)$$

is an exact $100(1 - \alpha)$ percent confidence interval for μ .

However, in most real problems, rarely will anyone tell us the value of σ^2 . That is, it is almost always the case that the population variance σ^2 is an unknown parameter. One might think to try using S as a point estimator for σ and substituting it into the confidence interval formula above. This is certainly not illogical, but, the sample standard deviation S is not an unbiased estimator for σ . Thus, if the sample size is small, there is no guarantee that sample standard deviation S will be *close* to the population standard deviation σ (it likely will be *close* if the sample size n is large). Furthermore, when the sample size n is small, the bias and variability associated with S (as an estimator of σ) could be large. To address this difficulty, we recall the following result.

Suppose that $Y_1, Y_2, \dots, Y_n$ is an *iid* sample from a $N(\mu, \sigma^2)$ population. From past results, we know that

$$T = \frac{\bar{Y} - \mu}{S/\sqrt{n}} \sim t(n-1).$$

Note that since the sampling distribution of T is free of all unknown parameters, T a pivotal quantity. So, just like before, we can use this fact to derive an exact $100(1-\alpha)$ percent confidence interval for μ . The derivation is shown below. Let $t_{n-1, \alpha/2}$ denote the upper $\alpha/2$ quantile of the $t(n-1)$ distribution.

$$\begin{aligned} 1 - \alpha &= P[-t_{n-1, \alpha/2} \leq T \leq t_{n-1, \alpha/2}] \\ &= P[-t_{n-1, \alpha/2} \leq \frac{\bar{Y} - \mu}{S/\sqrt{n}} \leq t_{n-1, \alpha/2}] \\ &= P[-t_{n-1, \alpha/2} \times (S/\sqrt{n}) \leq \bar{Y} - \mu \leq t_{n-1, \alpha/2} \times (S/\sqrt{n})] \\ &= P[t_{n-1, \alpha/2} \times (S/\sqrt{n}) > \mu - \bar{Y} > -t_{n-1, \alpha/2} \times (S/\sqrt{n})] \\ &= P[\bar{Y} - t_{n-1, \alpha/2} \times (S/\sqrt{n}) \leq \mu \leq \bar{Y} + t_{n-1, \alpha/2} \times (S/\sqrt{n})] \end{aligned}$$

This argument shows that

$$\bar{Y} \pm t_{n-1, \alpha/2} \times \left(\frac{S}{\sqrt{n}} \right)$$

is an exact $100(1-\alpha)$ percent confidence interval for the population mean μ . This interval is *exact* only if the underlying probability distribution is normal.

Example 3.5.1

In an agricultural experiment, a random sample of $n = 10$ plots produces the yields below (measured in kg per plot). From past studies, it has been observed that plot yields vary according to a normal distribution. The goal is to write a 95% confidence interval for μ , the population mean yield. Here are the sample yields:

23.2 20.1 18.8 19.3 24.6 27.1 33.7 24.7 32.4 17.3

SOLUTION

From these data, we compute $\bar{y} = 24.1$ and $s = 5.6$. Also, with $n = 10$, the degrees of freedom is $n - 1 = 9$, and $t_{9,0.025} = 2.262$. Hence, the 95% confidence interval is

$$24.1 \pm 2.262 \times \left(\frac{5.6}{\sqrt{10}} \right) \implies (20.1, 28.1)$$

Thus, based on these data, we are 95% confident that the population mean yield μ is between 20.1 and 28.1 kg/plot.

Difference of two population means

Suppose we have two independent random samples $Y_{11}, Y_{12}, \dots, Y_{1n_1}$ from $N(\mu_1, \sigma_1^2)$, and $Y_{21}, Y_{22}, \dots, Y_{2n_2}$ from a $N(\mu_2, \sigma_2^2)$. and that we would like to construct a $100(1 - \alpha)$ percent confidence interval for the difference of population means $\mu_1 - \mu_2$.

Recall that

$$\begin{aligned}\bar{Y}_1 &\sim N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \\ \bar{Y}_2 &\sim N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)\end{aligned}$$

By straightforward calculation, it follows that

$$\bar{Y}_1 - \bar{Y}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Standardizing, we get

$$Z = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

Also recall that

$$\begin{aligned}\frac{(n_1 - 1)S_1^2}{\sigma_1^2} &\sim \chi^2(n_1 - 1) \\ \frac{(n_2 - 1)S_2^2}{\sigma_2^2} &\sim \chi^2(n_2 - 1)\end{aligned}$$

It follows that

$$\frac{(n_1 - 1)S_1^2}{\sigma_1^2} + \frac{(n_2 - 1)S_2^2}{\sigma_2^2} \sim \chi^2(n_1 + n_2 - 2)$$

The population variances are **nuisance** parameters in the sense that they are not the parameters of interest here. Still, they have to be estimated. We want to write a confidence interval for $\mu_1 - \mu_2$, but exactly how this interval is constructed depends on the true values of σ_1^2 and σ_2^2 . In particular, we consider two cases:

1. Equal variances: $\sigma_1^2 = \sigma_2^2 = \sigma^2$
2. Unequal variances: $\sigma_1^2 \neq \sigma_2^2$

When $\sigma_1^2 = \sigma_2^2 = \sigma^2$, we have

$$Z = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0, 1)$$

and

$$\frac{(n_1 - 1)S_1^2}{\sigma_1^2} + \frac{(n_2 - 1)S_2^2}{\sigma_2^2} = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{\sigma^2} \sim \chi^2(n_1 + n_2 - 2)$$

Thus,

$$T = \frac{\frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}}{\sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{\sigma^2} / (n_1 + n_2 - 2)}} \sim t(n_1 + n_2 - 2)$$

Algebraically, the last expression reduces to

$$T = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2)$$

where

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

is the pooled sample variance estimator of the common population variance σ^2 .

Notice that T has a sampling distribution that is free of all unknown parameters, thus, it is a pivotal quantity. We can use this fact to construct a $100(1 - \alpha)$ percent confidence interval for mean difference $\mu_1 - \mu_2$. In particular, because $T \sim t(n_1 + n_2 - 2)$, we can find the value $t_{n_1+n_2-2, \alpha/2}$ that satisfies

$$P[-t_{n_1+n_2-2, \alpha/2} \leq T \leq t_{n_1+n_2-2, \alpha/2}] = 1 - \alpha$$

Substituting T into the last expression and performing the usual algebraic manipulations (VERIFY!), we can conclude that

$$(\bar{Y}_1 - \bar{Y}_2) \pm t_{n_1+n_2-2, \alpha/2} \times S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

is an exact $100(1 - \alpha)$ percent confidence interval for the mean difference $\mu_1 - \mu_2$.

Example 3.5.2

In the vicinity of a nuclear power plant, marine biologists at the EPA would like to determine whether there is a difference between the mean weight in two species of a certain fish. To do this, they will construct a 90% confidence interval for the mean difference $\mu_1 - \mu_2$. Two independent random samples were taken, and here are the recorded weights (in ounces):

Species 1 : 29.9, 11.4, 25.3, 16.5, 21.1

Species 2 : 26.6, 23.7, 28.5, 14.2, 17.9, 24.3

SOLUTION

Out of necessity, the scientists assume that each sample arises from a normal distribution with $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (i.e., they assume a common population variance). Here, we have $n_1 = 5, n_2 = 6, n_1 + n_2 - 2 = 9$, and $t_{9,0.05} = 1.833$. Straightforward computations show that $\bar{y}_1 = 20.84, s_1^2 = 52.50, \bar{y}_2 = 22.53, s_2^2 = 29.51$, and that

$$s_p^2 = \frac{(5-1)52.50 + (6-1)29.51}{5+6-2} = 39.73.$$

Thus, the 90% confidence interval for $\mu_1 - \mu_2$, based on these data, is given by

$$(20.84 - 22.53) \pm 1.833 \times \sqrt{39.73(1/5 + 1/6)} \implies (-8.69, 5.31)$$

Therefore, we are 90% confident that the mean difference $\mu_1 - \mu_2$ is between -8.69 and 5.31 ounces. Since this interval includes 0, this analysis does not suggest that the mean species weights, μ_1 and μ_2 , are truly different.

When $\sigma_1^2 \neq \sigma_2^2$, the problem of constructing a $100(1 - \alpha)$ percent confidence interval for $\mu_1 - \mu_2$ becomes markedly more difficult. The reason why this is true stems from the fact that there is no *obvious* pivotal quantity to construct (go back to the equal-variance case and see how this assumption simplified the derivation). However, in this situation, we can still write an approximate confidence interval for $\mu_1 - \mu_2$. This interval is given by

$$(\bar{Y}_1 - \bar{Y}_2) \pm t_{\nu, \alpha/2} \times \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

where the degree of freedom parameter ν is approximated by

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2-1}}$$

This formula for ν is called **Satterwaite's formula**. The derivation of this interval is left to another day.

Robustness of the t procedures

In the derivation of the one and two-sample confidence intervals for normal means (based on the t distribution), we have explicitly assumed that the underlying population distribution(s) was/were normal. Under the normality assumption,

$$\bar{Y} \pm t_{n-1, \alpha/2} \times \left(\frac{S}{\sqrt{n}} \right)$$

is an exact $100(1 - \alpha)$ percent confidence interval for the population mean μ . Under the normal, independent sample, and constant variance assumptions,

$$(\bar{Y}_1 - \bar{Y}_2) \pm t_{n_1+n_2-2, \alpha/2} \times S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

is an exact $100(1 - \alpha)$ percent confidence interval for the mean difference $\mu_1 - \mu_2$. Of course, the natural question arises: **What if the data are NOT normally distributed?**

By definition, a statistical inference procedure (like constructing a confidence interval) is said to be robust if the quality of the procedure is not affected by a departure from the assumptions made.

The t confidence interval procedures are based on the population distribution being normal. However, these procedures are fairly robust to departures from normality; that is, even if the population distribution(s) is/are nonnormal, we can still use the t procedures and get approximate results. The following guidelines are common:

1. $n < 15$: Use t procedures only if the population distribution appears normal and there are no outliers.
2. $15 \leq n \leq 40$: Be careful about using t procedures if there is strong skewness and/or outliers present.
3. $n > 40$: t procedures should be fine regardless of the shape of the population distribution.

These are just guidelines and should not be taken as *truth*. Of course, if we know the distribution of $Y_1, Y_2, \dots, Y_n$ (e.g., Poisson, exponential, etc.), then we might be able to derive an exact $100(1 - \alpha)$ percent confidence interval for the mean directly by finding a suitable pivotal quantity. In such cases, it may be better to avoid the t procedures altogether.

Confidence intervals for variances

In many experimental settings, the researcher is concerned not with the mean of the underlying population, but with the population variance σ^2 instead. For example, in a laboratory

setting, chemists might wish to estimate the variability associated with a measurement system (e.g., scale, caliper, etc.) or to estimate the unit-to-unit variation of vitamin tablets. In large-scale field trials, agronomists are often likely to compare variability levels for different cultivars or genetically-altered varieties. In clinical trials, the FDA is often concerned whether or not there is significant variation among various clinic sites. We examine the one and two-sample problems here.

One population variance

Suppose $Y_1, Y_2, \dots, Y_n$ is an iid sample from a $N(\mu, \sigma^2)$ distribution. In this case, we know

$$Q = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

Because Q has a distribution that is free of all unknown parameters, it is a pivot. We will use this pivot to derive an exact $100(1-\alpha)$ percent confidence interval for σ^2 .

Let $\chi_{n-1, \alpha/2}^2$ denote the upper $\alpha/2$ quantile and $\chi_{n-1, 1-\alpha/2}^2$ denote the lower $\alpha/2$ quantile of the $\chi^2(n-1)$ distribution. Then,

$$\begin{aligned} 1 - \alpha &= P[\chi_{n-1, 1-\alpha/2}^2 \leq Q \leq \chi_{n-1, \alpha/2}^2] \\ &= P[\chi_{n-1, 1-\alpha/2}^2 \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi_{n-1, \alpha/2}^2] \\ &= P\left[\frac{1}{\chi_{n-1, 1-\alpha/2}^2} > \frac{\sigma^2}{(n-1)S^2} > \frac{1}{\chi_{n-1, \alpha/2}^2}\right] \\ &= P\left[\frac{(n-1)S^2}{\chi_{n-1, 1-\alpha/2}^2} > \sigma^2 > \frac{(n-1)S^2}{\chi_{n-1, \alpha/2}^2}\right] \\ &= P\left[\frac{(n-1)S^2}{\chi_{n-1, \alpha/2}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{n-1, 1-\alpha/2}^2}\right] \end{aligned}$$

This argument shows that

$$\left[\frac{(n-1)S^2}{\chi_{n-1, \alpha/2}^2}, \frac{(n-1)S^2}{\chi_{n-1, 1-\alpha/2}^2} \right]$$

is an exact $100(1-\alpha)$ percent confidence interval for the population variance σ^2 .

Example 3.5.3

Entomologists studying the bee species *Euglossa mandibularis* Friese measure the wing-stroke frequency for $n = 4$ bees for a fixed time. The data are

235, 225, 190, 188

Assuming that these data are an *iid* sample from a $N(\mu, \sigma^2)$ distribution, Find a 90% confidence interval for σ^2 .

SOLUTION

Here, $n = 4$, $s^2 = 577.6667$, and $\alpha = 0.10$, so we need $\chi_{3,0.95}^2 = 0.351846$ and $\chi_{3,0.05}^2 = 7.81473$. The 90% confidence interval is

$$\left[\frac{3(577.6667)}{7.81473}, \frac{3(577.6667)}{0.351846} \right] \implies (221.76, 4925.45)$$

That is, we are 90% confident that the true population variance σ^2 is between 221.76 and 4925.45. Or, the true population standard deviation σ is between 14.89 and 70.18. Of course, both of these intervals are quite wide, but remember that $n = 4$, so we shouldn't expect notably precise intervals.

Ratio of two variances

Suppose that we have two independent samples:

$$\text{Sample 1 : } Y_{11}, Y_{12}, \dots, Y_{1n_1} \sim N(\mu_1, \sigma_1^2)$$

$$\text{Sample 2 : } Y_{21}, Y_{22}, \dots, Y_{2n_2} \sim N(\mu_2, \sigma_2^2)$$

and that we would like to construct a $100(1 - \alpha)$ percent confidence interval for $\theta = \sigma_2^2/\sigma_1^2$, the ratio of the population variances. Under these model assumptions, we know that

$$\frac{(n_1 - 1)S_1^2}{\sigma_1^2} \sim \chi^2(n_1 - 1), \text{ and}$$
$$\frac{(n_2 - 1)S_2^2}{\sigma_2^2} \sim \chi^2(n_2 - 1)$$

and that these two quantities are independent. It follows that

$$F = \frac{\frac{(n_1-1)S_1^2}{\sigma_1^2}/(n_1-1)}{\frac{(n_2-1)S_2^2}{\sigma_2^2}/(n_2-1)} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1)$$

Because F has a distribution that is free of all unknown parameters, it is a pivot, and we can use it to derive a $100(1 - \alpha)$ percent confidence interval for $\theta = \sigma_2^2/\sigma_1^2$.

Let $F_{n_1-1, n_2-1, \alpha/2}$ denote the upper $\alpha/2$ quantile and $F_{n_1-1, n_2-1, 1-\alpha/2}$ denote the lower $\alpha/2$ quantile of the $F(n_1 - 1, n_2 - 1)$ distribution. Then,

$$\begin{aligned}
1 - \alpha &= P[F_{n_1-1, n_2-1, 1-\alpha/2} \leq F \leq F_{n_1-1, n_2-1, \alpha/2}] \\
&= P\left[F_{n_1-1, n_2-1, 1-\alpha/2} \leq \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \leq F_{n_1-1, n_2-1, \alpha/2}\right] \\
&= P\left[\frac{S_2^2}{S_1^2} \times F_{n_1-1, n_2-1, 1-\alpha/2} \leq \frac{\sigma_2^2}{\sigma_1^2} \leq \frac{S_2^2}{S_1^2} \times F_{n_1-1, n_2-1, \alpha/2}\right]
\end{aligned}$$

This argument shows that

$$\left(\frac{S_2^2}{S_1^2} \times F_{n_1-1, n_2-1, 1-\alpha/2}, \frac{S_2^2}{S_1^2} \times F_{n_1-1, n_2-1, \alpha/2}\right)$$

is a $100(1 - \alpha)$ percent confidence interval for $\theta = \sigma_2^2/\sigma_1^2$.

Example 3.5.4

Snout beetles cause millions of dollars worth of damage each year to cotton crops. Two different chemical treatments are used to control this beetle population using 13 randomly selected plots. Below are the percentages of cotton plants with beetle damage (after treatment) for the plots:

Treatment 1 : 22.3, 19.5, 18.6, 24.3, 19.9, 20.4

Treatment 2 : 9.8, 12.3, 16.2, 14.1, 15.3, 10.8, 18.3

Under normality, and assuming that these two samples are independent, find a 95% confidence interval for $\theta = \sigma_2^2/\sigma_1^2$, the ratio of the two treatment variances.

SOLUTION

Here, $n_1 = 6, n_2 = 7, s_1^2 = 4.40, s_2^2 = 9.27$, and $\alpha = 0.05$, so that $F_{5,6,0.025} = 5.99$ and $F_{5,6,0.975} = 0.14$.

Thus, a 95% confidence interval for $\theta = \sigma_2^2/\sigma_1^2$ is given by

$$\left(\frac{9.27}{4.40} \times 0.14, \frac{9.27}{4.40} \times 5.99\right) \implies (0.29, 12.62)$$

Thus, we are 95% confident that the ratio of variances $\theta = \sigma_2^2/\sigma_1^2$ is between 0.29 and 12.62. Since this interval includes 1, we can not conclude that the two treatment variances are significantly different.

REMARK

Unlike the t confidence intervals for means, the confidence interval procedures for one and two population variances are not robust to departures from normality. Thus, one who uses these confidence intervals is placing strong faith in the underlying normality assumption.